

Title: Sales Performance Analysis using Tableau

1. Objective

The objective of this experiment is to analyze the sales performance of a retail company over multiple years using Tableau. The goal is to identify trends, patterns, and insights by visualizing data using different types of charts.

2. Dataset Description

The dataset used in this analysis is a retail sales dataset containing the following fields:

- **Category** – Type of product (Electronics, Furniture, Office Supplies)
- **Customer Name** – Name of the customer
- **Order Date** – Date of purchase
- **Order ID** – Unique transaction identifier
- **Payment Mode** – Payment method used
- **State and City** – Location of the customer
- **Sub-Category** – Detailed product classification
- **Year-Month** – Time-based grouping
- **Amount** – Sales value
- **Profit** – Profit earned
- **Quantity** – Number of items sold

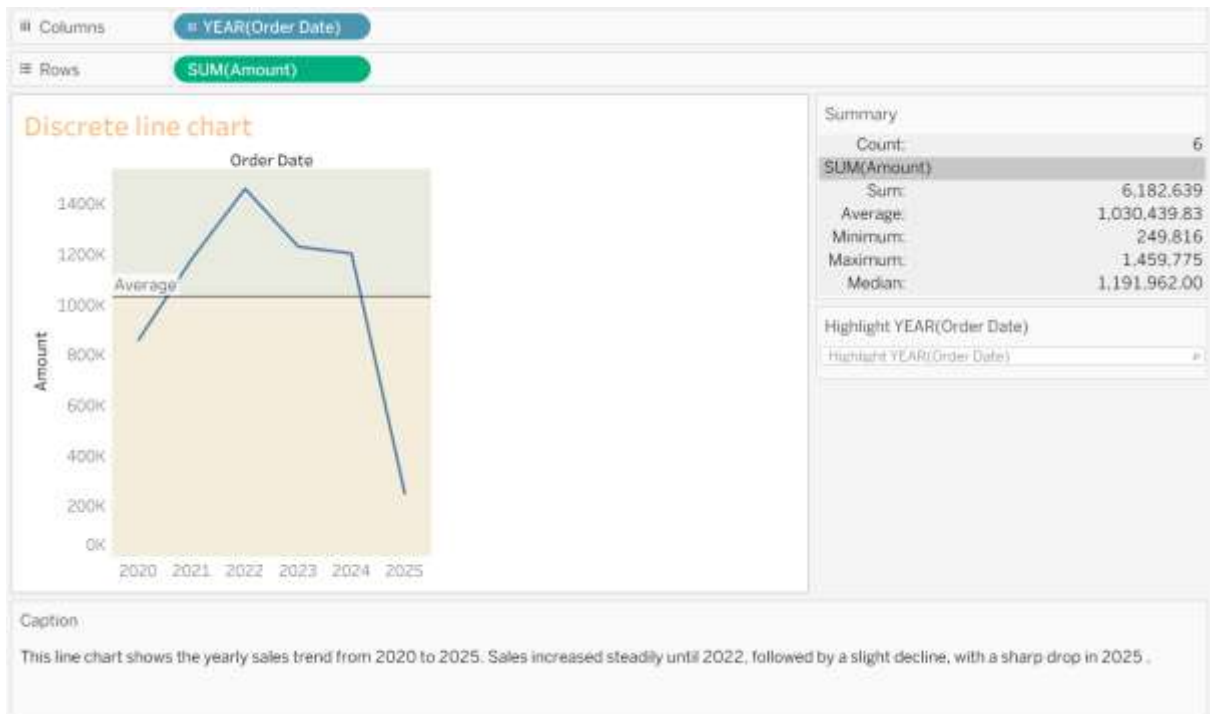
3. Tools Used

- Tableau (Data Visualization Tool)

4. Visualizations and Analysis

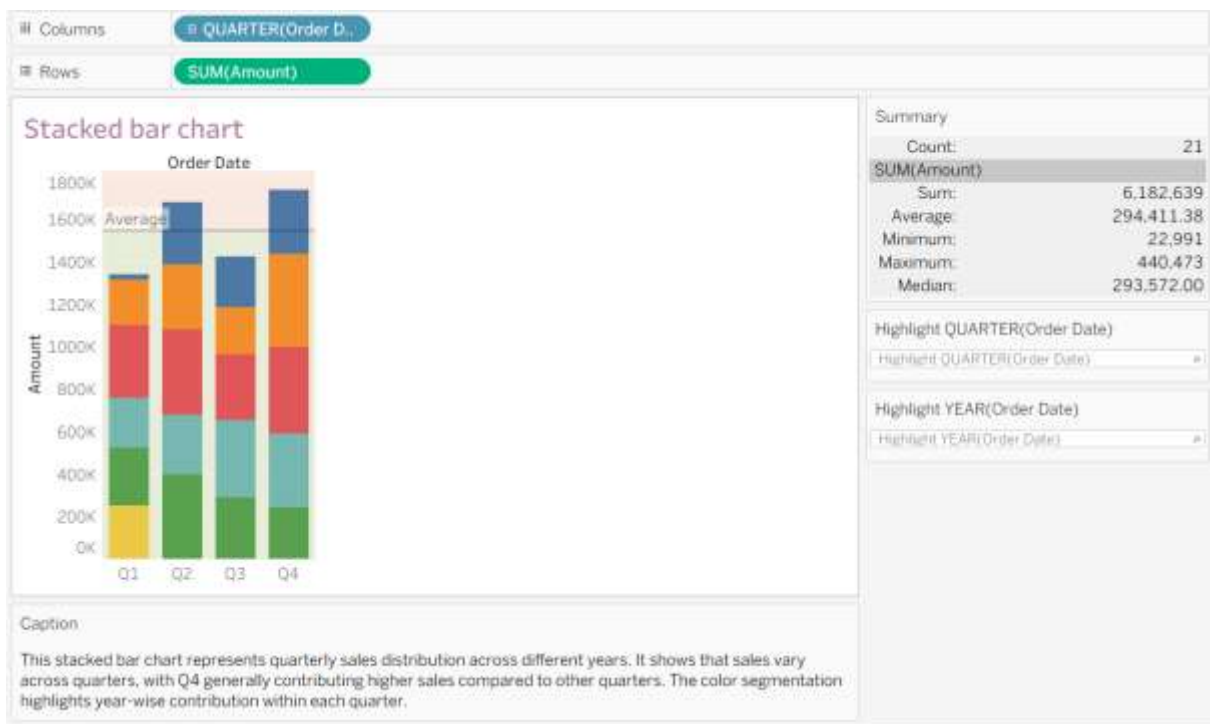
4.1 Line Chart (Yearly Sales Trend)

- Displays sales over different years using Order Date and Amount
- Shows increasing trend from 2020 to 2022
- Peak sales observed in 2022
- A decline is observed after 2022, with a sharp drop in 2025



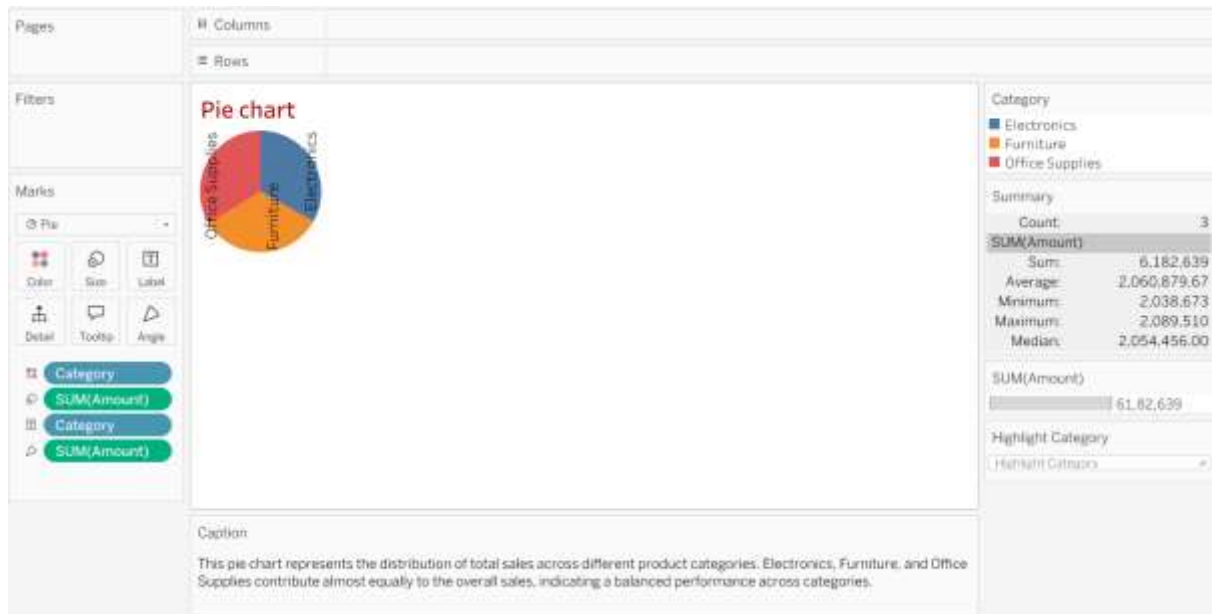
4.2 Bar Chart (Year-wise Comparison)

- Compares total sales across years
- Clearly highlights highest and lowest performing years
- Useful for quick comparison



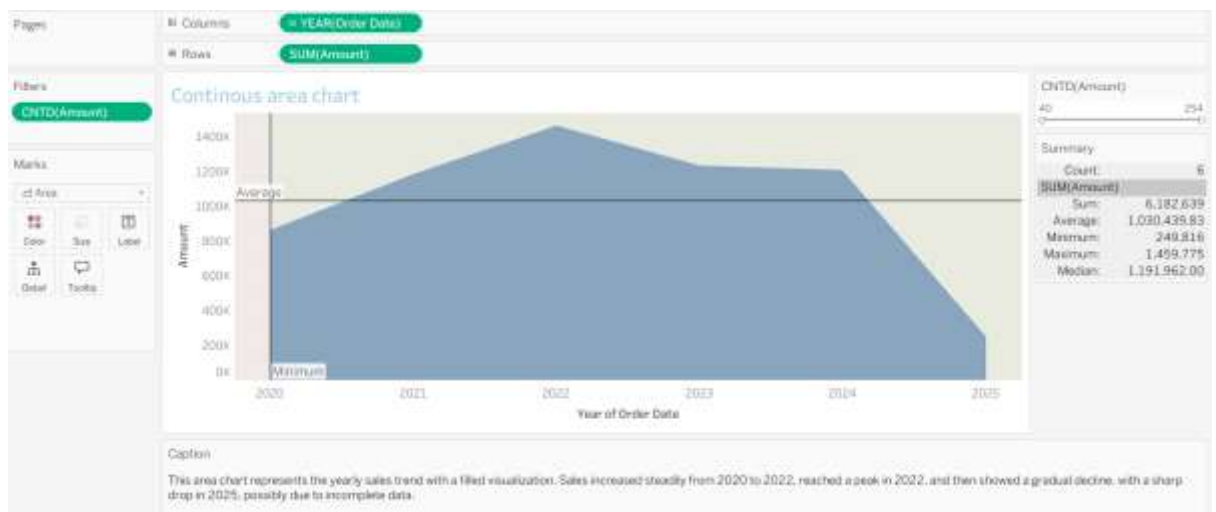
4.3 Pie Chart (Category-wise Distribution)

- Represents contribution of each category to total sales
- Electronics, Furniture, and Office Supplies are compared
- Helps identify dominant categories



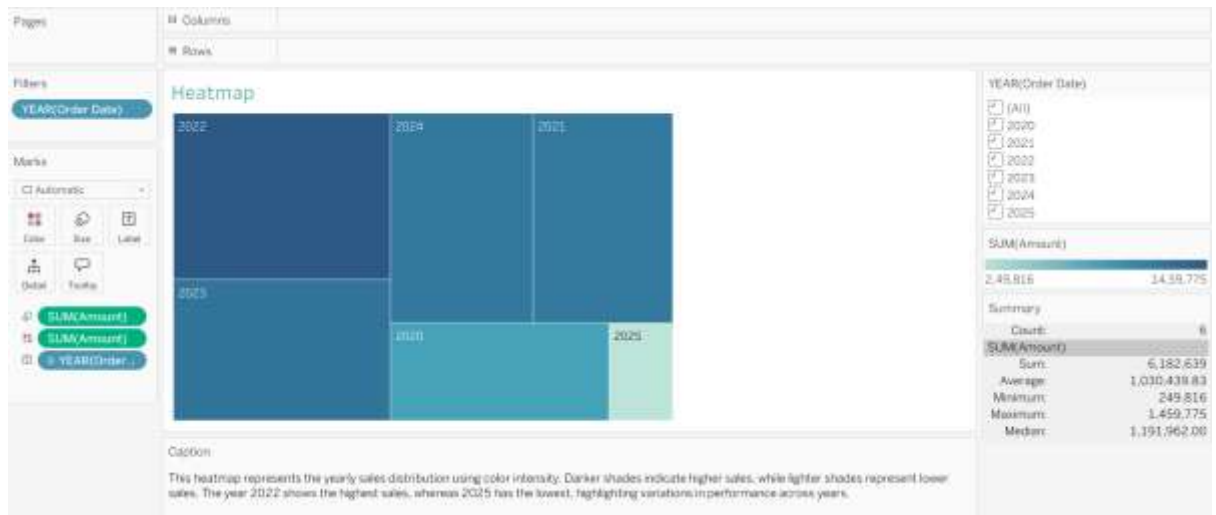
4.4 Area Chart (Sales Trend Visualization)

- Shows sales trend with filled area
- Highlights growth and decline patterns
- Provides better visual impact than line chart



4.5 Heatmap (Year-wise Sales Intensity)

- A heatmap was used to visualize sales using color intensity
- Darker colours represent higher sales values
- The year 2022 shows the highest sales performance
- The year 2025 shows the lowest sales
- It helps quickly identify variations and patterns in sales data



5. Key Insights

- Sales increased significantly from 2020 to 2022
- 2022 is the highest performing year
- Slight decline observed in later years
- Categories contribute almost equally to total sales
- Visualization improves understanding of trends and patterns

6. Conclusion

The analysis demonstrates how Tableau can be effectively used to visualize and interpret sales data. Different visualization techniques provide meaningful insights, enabling better understanding of business performance and supporting data-driven decision-making.